



# Advances in retinal microaneurysms detection, segmentation and datasets for the diagnosis of diabetic retinopathy: a systematic literature review

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## Abstract

Microaneurysms (MAs) are small, circular, red lesions typically observed in the initial stage of Diabetic Retinopathy (DR). DR is a degenerative eye condition resulting from diabetes mellitus that can lead to vision loss if not detected and treated early. Since MAs are one of the first visible signs of DR, detecting them can lead to early diagnosis and treatment of DR and potentially prevent severe vision loss. In this survey, we conduct a thorough review and analyze the published literature on automated detection and segmentation of MAs, with the aim of bolstering ophthalmologists capabilities in the early screening and management of DR. Section 2 provides an overview of publicly available datasets that include image-level and pixel-level annotations of MAs. Benchmarking plays a vital role in the precise assessment of the efficacy of early DR detection systems. Section 3 presents common metrics used to benchmark the performance of MA recognition technologies. We categorize MA recognition methodologies into two principal groups: those based on image processing and those leveraging deep learning techniques. Sections 4 and 5 meticulously enumerate the cutting-edge techniques for MA detection and segmentation. Our discussion also delves into recent advancements and explores prospective future directions in this field. In conclusion, the evolution of machine learning and computer vision has enabled automated MA recognition methods to exhibit considerable potential, marking a significant stride forward in the fight against DR-related vision loss.

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## 1 Introduction

Diabetes Mellitus (DM) is a worldwide epidemic responsible for numerous health complications and fatalities [1]. It is a chronic medical condition characterized by elevated blood glucose levels and is rapidly emerging as one of the foremost global health challenges of the 21st century. It has a profound impact on blood vessels and nerves, leading to substantial damage to essential organs such as the brain, eyes, kidneys and heart. In its 2016 report on the worldwide prevalence of diabetes, the World Health Organization noted that approximately 463 million individuals aged 20 to 79 were affected by the condition. This number is expected to rise to 700 million by 2045 if current trends persist [2].

Diabetic Retinopathy (DR) is a major complication for those suffering from DM, affecting the blood vessels in the retina and leading to the accumulation of fluid. At present, approximately 30% of individuals with DM experience DR. Without appropriate treatment, DR can progress to its most severe stage, known as proliferative DR (PDR), which can cause vision loss in working-age adults [3].

According to the American Academy of Ophthalmology, DR is classified into two main categories: proliferative DR (PDR) and non-proliferative DR (NPDR), based on the presence or absence of specific retinal characteristics [4]. The term “proliferative” indicates the presence of Neo-Vascularization (NV), which is the abnormal growth of blood vessels in the retina, whereas “non-proliferative” refers to its absence.

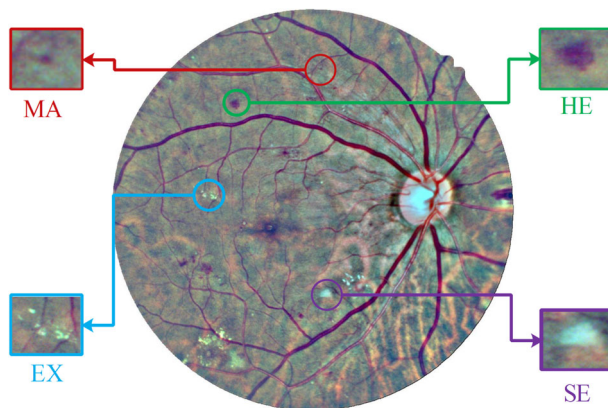
The classification of NPDR into four stages; mild, moderate, severe, and very severe is based on these lesion types [5, 6]. Table 1 offers further details regarding the classification of diabetic retinopathy, utilizing experimental clinical data from the retina [6, 7]. NPDR includes a range of retinal symptoms, such as Hemorrhage (HE), Microaneurysm (MA), Hard Exudate (EX), and Soft Exudate (SE), as illustrated in Fig. 1. The categorization of NPDR can be further classified into four stages, namely mild, moderate, severe and very severe, according to the International Clinical Diabetic Retinopathy [8] (Table 1). In fundus images, MAs appear as small, red, isolated spots in close proximity to small blood vessels and are indicative of early DR. As the disease progresses, more MAs can be observed near these vessels, along with areas of HEs and venous distortion, all of which contribute to visual impairment in the affected individuals.

Detecting MA is crucial for preventing blindness by enabling cost-effective healthcare strategies through the early detection of DR. However, identifying MAs within a fundus image is a significant challenge. As a result, ophthalmologists may not be able to recognize the majority of MAs quickly when screening a large number of patients. Therefore, the adoption of automated computer-aided diagnostic systems is vital for improving efficiency and reducing expenses in mass screening programs, as emphasized in the study by Dashtbozorg et al. (2018) [9].

The main contribution of this paper is an in-depth analysis and assessment of diverse approaches to identify MA. Figure 2 shows a mind map of our review paper, which pertains to MA recognition in retinal fundus images. The mind map is color-coded to denote various sections of the research, including the introduction, methods for MA detection, segmentation techniques, databases, evaluation metrics, discussion, and conclusion. This paper is organized as follows:

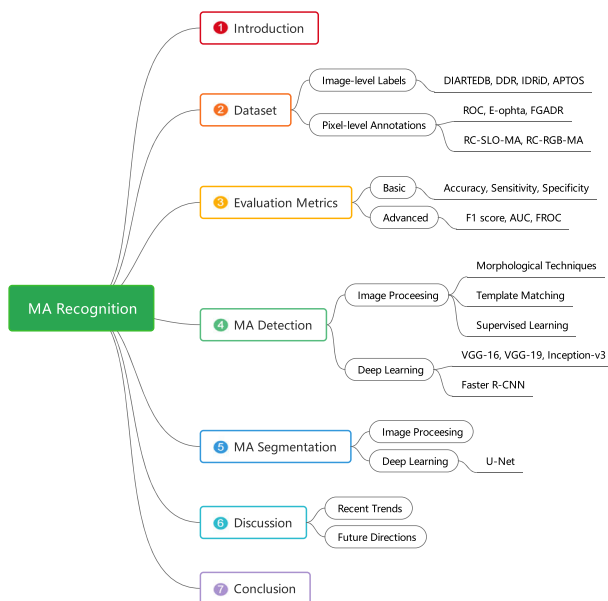
Table 1 Categorization of DR-based clinical outcomes of diabetic retinopathy

| Grade | Category         | DR based clinical findings                                                                                                                                                                                     |                        |
|-------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|       |                  | Sufficient                                                                                                                                                                                                     | Indirect               |
| DR0   | No DR            | No abnormalities                                                                                                                                                                                               | -                      |
| DR1   | Mild NPDR        | MAAs detected                                                                                                                                                                                                  | -                      |
| DR2   | Moderate NPDR    | Contains two or more of the qualities listed below: MAAs, HEs, EXs less than severe NPDR                                                                                                                       | EXs                    |
| DR3   | Severe NPDR      | 20 HEs individually separated into four quadrants “or” intraretinal microvascular abnormalities in one quadrant “or” venous beading in two quadrants.<br>Any two severe NPDR features present at the same time | SEs                    |
| DR4   | Very severe NPDR | Any two severe NPDR features present at the same time                                                                                                                                                          | -                      |
| DR5   | PDR              | It implements one of the following:<br>NV, vitreous haemorrhage, preretinal haemorrhage                                                                                                                        | fibrous proliferation. |



**Fig. 1** A color fundus image with retinal lesions. The cropped image patches show the existence of hemorrhage, microaneurysm, soft exudate, and hard exudate

- Section 2 provides a comprehensive survey of the fundus image datasets employed for benchmarking MA detection and segmentation methods. It outlines and discusses the datasets that are currently available for benchmarking purposes.
- Section 3 details performance metrics of the study, which are elaborated upon and discussed comprehensively. This section also provides a concise explanation of the key performance indicators employed in early diagnosis systems.
- Section 4 provides a comprehensive review of existing methods for the identification of MAs in retinal images. This section presents an overview of the current identification techniques, exploring both machine learning and image processing strategies. Addition-



**Fig. 2** The structure of our review paper on MA recognition: we categorize automated MA identification techniques into two main tracks - detection and segmentation

ally, it offers a critical assessment of the advantages and drawbacks associated with these methods.

- Section 5 presents an in-depth analysis of segmentation techniques used for MA detection in fundus images. It categorizes the methods into image-processing-based and machine learning-based segmentation techniques, providing a thorough evaluation of their effectiveness and limitations.
- Section 6 discusses recent advancements and future prospects in the recognition of MAs, highlighting the cutting-edge advancements in this area.
- Finally, Section 7 concludes the paper by summarizing the main findings and conclusions drawn from our research, outlining the significant insights and contributions to the field.

## 2 Datasets

Several retinal datasets that can be used for research and scientific studies are publicly available. All public datasets aim to provide researchers with an explicit platform to authenticate their algorithms. These databases facilitate easy comparison and valid inference after considering suitable environments. They make comparing results easier and drawing accurate conclusions by ensuring similar testing conditions and procedures. The existing retinal fundus images dataset consists of two categories: image-level labels and pixel-level annotations (for MAs detection).

### 2.1 Image-level labels

Images-level labeling is the process of an entire image label, and one provides us with more descriptive information rather than labeling individual elements within an image. The primary objective of image-level labeling in MA involves identifying and marking the presence of MA with a colour fundus image. It focuses on categorizing or describing the content of the entire image based on its overall context or characteristics. For instance, in medical imaging like retinal fundus images, image-level labeling might involve tagging an entire image as having certain features (e.g., the presence of specific abnormalities like MAs) rather than identifying and marking each instance of that feature within the image. The DIARTEDB [102], IDRiD [103], DDR [104], and APTOS 2019 Kaggle datasets [105] use image-level labels to detect MAs in retinal fundus images. Table 2 details existing fundus image datasets published recently and publicly available for researchers.

#### 2.1.1 DIARTEDB

The DIARTEDB dataset [102] has been publicly available since 2006, obtained from the Kuopio university hospital in Finland. The dataset contains 130 eye color fundus images and is taken by a digital fundus camera. The resolution of images is  $[1500 \times 1152]$  at 50-degree Field of View (FOV). Total no of 130 Images, 20 images are normal, but the remaining 110 images have diabetic retinopathy indicators.

#### 2.1.2 IDRiD

The Indian Diabetic Retinopathy Image Dataset (IDRiD) [103] is a publicly available dataset obtained from the Eye Clinic Nanded in India and organized into two further subgroups:

Table 2 Comparison between existing fundus image datasets

| Ref.                   | Database   | Year | Total Images | Healthy | images with DR | Image Modality | Resolution (height × width)              | FOV | Classification Task                                                                    |
|------------------------|------------|------|--------------|---------|----------------|----------------|------------------------------------------|-----|----------------------------------------------------------------------------------------|
| Decencière et al [106] | Messidor-2 | 2008 | 1200         | 540     | 660            | CF             | 1440 × 960<br>2240 × 1488<br>2304 × 1536 | 45° | DR grading 0-3.<br>Pupil dilation - 800, without dilation - 400                        |
| Porwal et al [103]     | IDRiD      | 2018 | 516          | 168     | 348            | CF             | 4288 × 2848                              | 50° | DR and DME                                                                             |
| Li et al [104]         | DDR        | 2019 | 13673        | 6266    | 6256           | CF             | Different image size                     | 45° | Segmentation & Grading<br>Lesions such as:<br>MAs, Soft exudates, EX, Hemorrhages (H). |
| Sikder et al [105]     | APTOS 2019 | 2019 | 3662         | 1805    | 1857           | CF             | Different image size                     | —   | DR grading 0-4 (no, mild, moderate, severe, and proliferative)                         |

normal images, retinal images with DR and Diabetic Macular Edema (DME) symptoms. The medical experts rated the rigour level of the DR scale from 0 to 4 and the risk score for diabetic macular edema on a scale from 0 (no DME) to 2 (severe DME) for a complete set of 516 images. The dataset holds 516 fundus images captured by Kowa VX-10a digital camera. The resolution of images is  $[4288 \times 2848]$ , at 50-degree FOV in JPG format, and each picture was about 800 KB in size. The image-level disease grading dataset is classified into 413 (80%) training and 103 (20%) test images. Then, 81 images with DR symptoms were described at the pixel level on typical DR lesions and the pixel level of the optic disc. IDRiD is a unique dataset that includes explanations of DME and DR pathology and large fundus structures, but IDRiD has few images.

### 2.1.3 DDR

The DDR dataset [104] contains 13,673 image-level annotated fundus images, of which 757 are annotated at the pixel and bounding box levels. In addition, DDR offers pixel and image-level annotations that enable the development of multitasking deep learning models. It offers bounding box and pixel annotations for EX, SE, HE, and MA. The DDR is the most significant image dataset that provides that facility. All images were acquired using 42 camera brands, such as Topcon TRC NW48, Topcon D7000, Canon CR 2, and Nikon D5200.

### 2.1.4 APTOS 2019 Kaggle

The APTOS 2019 BD (Asia Pacific Tele-Ophthalmology Society 2019 Blindness Detection) dataset [105] was acquired from Aravind Eye Hospital, India. The APTOS dataset is an open (Kaggle) competition. The dataset consists of 3662 fundus images. These RGB images are obtained through funds Photography. Trained physicians then labelled the samples and classified the person's blindness into many classes: no, mild, moderate, severe, and proliferative.

#### **Distribution of databases for image-level labels.**

#### **A. DIARETDB, DIARETDB1**

Dai et al. [16], Seoud et al. [23], Akram et al. [24], Ganjee et al. [25], Bharkad et al. [28] Ren et al. [29], Zhang et al. [33], Tavakoli et al. [34], Srivastava et al. [54], Dai et al. [42], Orlando et al. [43], Chudzik et al. [44], Khojasteh et al. [62], Harangi et al. [63], Roychowdhury et al. [107], Rocha et al. [108], Figueiredo et al. [109].

#### **B. IDRiD**

Kou et al. [46], Guo et al. [47], Garifullin et al [94], Huang et al. [110], Hasan et al. [111], Kalpana et al. [112], Shaik et al. [113], Erciyas et al. [114], Niu et al. [115], Chaudhary et al. [116].

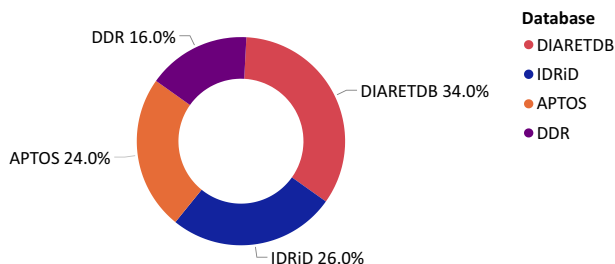
#### **C. DDR**

Hervella et al. [48], Guo et al. [47], Rocha et al. [108], Huang et al. [110], Guo et al. [117], Menaouer et al. [118], Starovoitov et al. [119].

#### **D. APTOS 2019**

Shaik et al. [113], Islam et al. [120], Zhang et al. [121], Salam et al. [122], Shaik et al. [113], Shaik et al. [123], Lahmar et al. [124], Alyoubi et al. [125], Nahiduzzaman et al. [126].

Figure 3 shows how often different image-level databases are used in research studies. It gives percentages for each database: DIARTEDB is used by 34% of the studies, IDRiD 26%, APTOS 24%, and DDR 16%, respectively.



**Fig. 3** The distribution of image-level databases used in different kinds of literature

## 2.2 Pixel-level Annotations

DR is an eye disease that can cause blindness by affecting the retina. Healthcare professionals rely on detailed pixel-level annotations in retinal images to diagnose and monitor DR accurately. Pixel-level annotations refer to the meticulous process of manually marking or labeling individual pixels within an image. In this method, each pixel in an image is examined and assigned specific annotations or labels based on certain characteristics or features present in that pixel. For instance, in medical imaging, such as retinal images, pixel-level annotations involve identifying and marking each pixel corresponding to features like MAs, blood vessels, or other abnormalities. This detailed annotation allows for precise analysis, aiding in the accurate diagnosis and tracking of conditions like DR by highlighting specific areas of interest within the image at the pixel level. Datasets like ROC, e-ophta, RC-RGB-MA, RC-SLO-MA, and FGADR use this labeling technique to segment MAs in retinal fundus images. Table 3 provides details about these fundus image datasets.

### 2.2.1 ROC

Retinopathy Online Challenge (ROC) was presented in 2009 by the University of Iowa [127]. The dataset offers a reference set for MA detection in color fundus images employed in the ROC for DR screening. This data set consisted of 100 digital photographs of the fundus containing MAs. The database is divided into 50 and 50 training image sets. All images are collected using the Topcon NW 100, Canon CR5-45NM, and Topcon NW200 at 45-degree FOV. The database has three image sizes;  $768 \times 576$ ,  $1058 \times 1061$ , and  $1389 \times 1383$ , and is compressed in JPEG format in the camera.

### 2.2.2 E-ophta

E-ophta [128] contains fundus images captured primarily for scientific research in the DR field. It includes two types, e-ophta MA and e-ophta EX associated with MA and EX annotations. The e-ophta EX contains 47 lesion images and 35 images of healthy individuals. In contrast, the e-ophta MA subset comprises 1306 MAs and 233 healthy photos. The camera compresses all images in JPEG format, and the dataset is 50 GB in size.

### 2.2.3 RC-SLO-MA

RC-RGB-MA [9] dataset was collected from the Netherlands Eindhoven University of Technology in the Netherlands. The dataset is compiled using the retina check framework. The

**Table 3** A comparison of existing fundus image datasets

| Ref.                   | Database     | Year | Total Images | Healthy | images with DR | Image Modality | Resolution (height × width)     | FOV | Task                   |
|------------------------|--------------|------|--------------|---------|----------------|----------------|---------------------------------|-----|------------------------|
| Niemeijer et al [127]  | ROC          | 2009 | 100          | 10      | 90             | CF             | 768 × 5761058 × 10611389 × 1383 | 45° | MA detection           |
| Decenciere et al [128] | E-ophtha- MA | 2013 | 381          | 233     | 148            | CF             | 2544 × 169 1440 × 960           |     | MA detection           |
| Dashthbozorg et al [9] | RC-RGB-MA    | 2018 | 250          |         |                | CF             | 2595 × 1944                     | 45° | MA detection           |
| Dashthbozorg et al [9] | RC-SLO-MA    | 2018 | 58           |         |                | CF             | 1024 × 1024                     | 45° | MA detection           |
| Zhou et al [129]       | FGADR        | 2020 | 2842         | 1000    | 1842           |                | Different image size            |     | Segmentation & Grading |

RGB fundus images were obtained using a fundus camera that was non-mydratic at  $2595 \times 1944$  resolution and a 45-degree FOV. Moreover, using the MA annotation tool, two professionals annotated MA on all photographs (RC-MAT).

## 2.2.4 RC-RGB-MA

RC-SLO-MA [9] is an RGB fundus images dataset acquired using Scanning Laser Ophthalmoscopy (SLO) technology. All dataset images are captured using an EasyScan camera along the 45-degree FOV. Moreover, the dataset includes 58 images at  $1024 \times 1024$  resolution.

## 2.2.5 FGADR

The FGADR dataset [129] has two subsets: Seg-set and Grade-set. The Seg-set includes 1842-pixel level annotated and grading label images. The pixel-level label consists of six categories of lesions, including MA, HE, EX, SE, intra-retinal microvascular abnormalities and NV. Three ophthalmologists annotate the grading labels. The second set, called Grade-set, contains 1000 graded-labelled images annotated by six ophthalmologists. The sample images of the dataset are shown in Fig. 4. We randomly selected two images, namely 0000\_1 and 0092\_1, respectively, from the FGADR dataset.

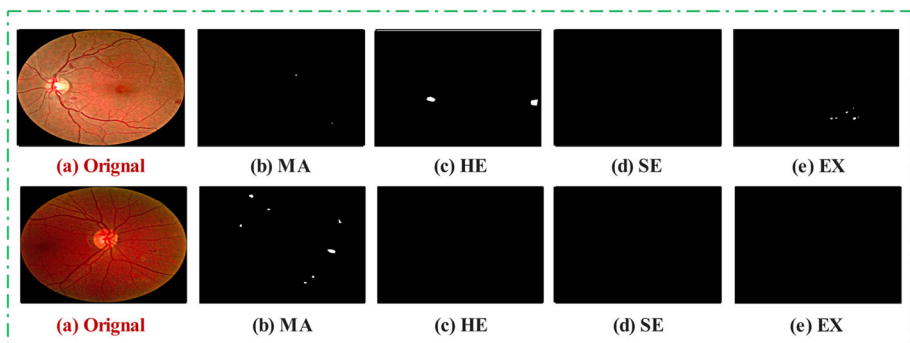
### Distribution of databases for Pixel-level annotations.

#### A. ROC Database

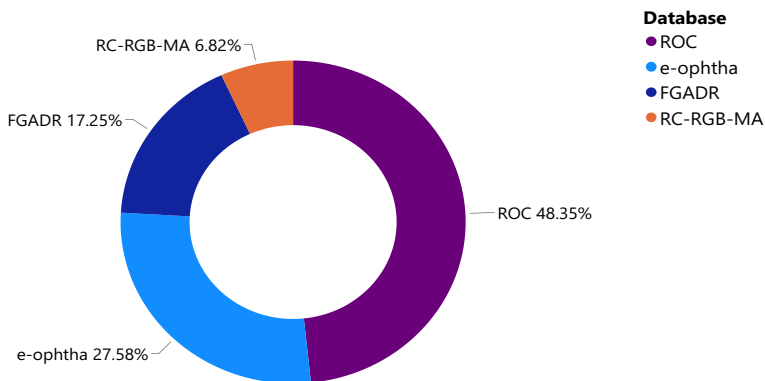
Dashtbozorg et al. [9], Melo et al. [10], Wang et al. [18], Quéllec et al. [20], Seoud et al. [23], Veiga et al. [26], Wu et al. [27], Ren et al. [29], Derwin et al. [30], Mazlan et al. [31], Derwin et al. [32], Zhang et al. [33], Xia et al. [52], Budak et al. [61], Chudzik et al. [44], Eftekhari et al. [45], Hervella et al. [48], Harangi et al. [63], Guo et al. [47], Mizutani et al. [37], Kar et al. [130], Giancardo et al. [131], Anta et al. [132], Lazar et al. [133], Sanchez et al. [134].

#### B. e-optha

Veiga et al. [26], Wu et al. [27], Dashtbozorg et al. [10], Melo et al. [12], Mazlan et al. [31], Zhang et al. [33], Ren et al. [29], Orlando et al. [43], chudzik et al. [44], Eftekhari [45], Hervella et al. [48], Khojasteh et al. [62], Harangi et al. [63], Kou et al. [46], Guo et al. [47].



**Fig. 4** Samples images of pixel-level annotations. The original photos and the ground truth for MA, HE, SE, and EX are displayed reading from left to right in this presentation



**Fig. 5** The distribution of pixel-level databases used in different kinds of literature

### C. RC-RGB-MA, RC-SLO-MA

Dashtbozorg et al. [9], Zhang et al. [33], Zhao et al. [135], Xia et al. [52].

### D. FGADR Zhou et al. [129], Ríos et al. [136], Zhou et al. [137], Gu et al. [138], Wang et al. [139], Chen et al. [140], Abdullah et al. [141].

In Fig. 5, you can see how frequently various pixel-level databases are used in research. It shows the percentages for each database: ROC is used in 48.35% of studies, e-ophta in 27.58%, FGADR in 17.25%, and RC-RGB-MA in 6.82%.

## 3 Evaluation metrics

Benchmarking is crucial for accurately evaluating how well early diagnosis systems for DR perform when using algorithms to detect and segment MAs. Different state-of-the-art methods were presented for this study and have adopted several metrics to analyze the same dataset. In addition, the study also used pixel-level and image-level-based evaluations. In pixel-level techniques, the segmented MA images are matched pixel-by-pixel with the ground truth mask Fig. 6. Table 4 briefly explains the key performance metrics used in early diagnosis systems.

Additionally, when employing image-level classification methods, the fundus images were categorized into normal and DR images. Different standard metrics are used to evaluate system performance for the above categories, such as specificity (False Positive Rate (FPR)),

**Fig. 6** It illustrates the connection between these principles and performance

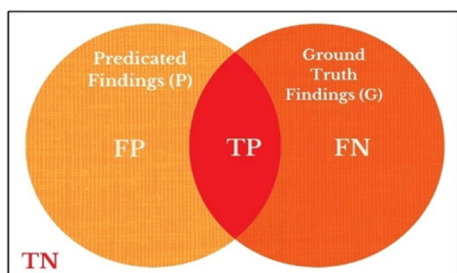


Table 4 Evaluation metrics

| Metric                                 | Formula                                              | Description                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Specificity/ FPR                       | $SP = \frac{TN}{(FN+TP)}$                            | The percentage of correctly identified findings.<br>The system is better if the value is high. FPR= (1-SP)                                                                                                     |
| Sensitivity / Recall / TPR             | $SE = \frac{TP}{(TP+FN)}$                            | The percentage of correctly identified non-MA.<br>The system is better if the value is high                                                                                                                    |
| Precision / Positive predictive value  | $PR = \frac{TP}{(TP+FP)}$                            | Illustrates how many of the labelled MAs are accurately identified. The system is better the greater the value.                                                                                                |
| F1 score / Dice similarity coefficient | $F1 = \frac{2 \times (TP)}{(2 \times TP + FP + FN)}$ | The harmonic meaning of precision and recall.<br>The higher value of the F1 score the system indicates better performance.                                                                                     |
| Accuracy                               | $ACC = \frac{(TP+TN)}{(TP+FP+FN+TN)}$                | The percentage of accurately predicted MA to total observed MA.                                                                                                                                                |
| AUC                                    |                                                      | The ROC curve is a probability curve (plotted as outlier sensitivity or sensitivity versus fall-out), and the AUC measures class separability. a higher AUC suggests better distinction between MA and non-MA. |
| FROC                                   |                                                      | The classifier's performance is shown by plotting the true positive rate against the false positive rate, with FROC analysis considering the number of MAs and accuracy.                                       |

**Table 5** Performance comparison of MA detection methods on Public Dataset

| Reference                | Database                 | ACC   | PR    | SE    | SP    | AUC   | F1    |
|--------------------------|--------------------------|-------|-------|-------|-------|-------|-------|
| Dai et al [16]           | DiaretDB1                | 0.961 | 0.997 |       |       |       | 0.934 |
| Orlando et al [43]       | E-Ophtha                 |       |       | 0.910 | 0.972 | 0.893 |       |
| Harangi et al [63]       | ROC, E-Ophtha, DiaretDB1 | 0.694 |       |       |       | 0.833 |       |
| Alyoubi et al [125]      | DDR                      | 0.890 |       | 0.985 | 0.993 | 0.970 |       |
| Akram et al [24]         | DiaretDB1                |       |       | 0.978 | 0.983 |       |       |
| Roychowdhury et al [107] | DiaretDB1                |       |       | 0.800 | 1.000 | 0.850 | 0.531 |
|                          | MESSIDOR                 |       |       |       |       |       |       |
| Ren et al [29]           | E-Ophtha                 |       |       | 0.961 | 0.821 |       |       |
| Rocha et al [108]        | DiaretDB1                |       |       | 0.900 | 0.600 |       |       |
| Agurto et al [142]       | MESSIDOR                 |       |       | 0.920 | 0.540 |       |       |
| Mizutani et al [37]      | ROC                      |       |       | 0.648 |       |       |       |
| Wang et al [143]         | DDR                      | 0.836 | 0.831 |       |       |       |       |
| Das et al [144]          | DiaretDB1                | 0.987 | 0.972 | 0.996 | 0.982 |       |       |
| Shankar et al [145]      | MESSIDOR                 | 0.992 |       | 0.996 | 0.982 |       |       |
| Shankar et al [146]      | MESSIDOR                 | 0.994 |       | 0.988 | 0.966 |       |       |

**Table 6** Performance comparisons of MA Detection approaches using Private Datasets

| Ref.                       | Database | images | SE    | SP    | AUC   | F1    |
|----------------------------|----------|--------|-------|-------|-------|-------|
| Burckhardt et al [147]     | Private  | 140    | 1.000 | 0.870 |       |       |
| Fleming et al [14]         | Private  | 1441   | 0.854 | 0.831 |       | 0.830 |
| Cree et al [148]           | Private  | 20     | 0.820 | 0.840 |       |       |
| Hänsen et al [149]         | Private  | NA     | 0.953 | 0.936 |       |       |
| Frame et al [36]           | Private  | 20     | 0.840 | 0.850 |       |       |
| Hipwell et al [150]        | Private  | 3783   | 0.810 | 0.930 |       |       |
| Sinthanayothin et al [151] | Private  | 30     | 0.775 | 0.887 |       |       |
| Usher et al [152]          | Private  | 773    | 0.951 | 0.463 |       |       |
| Niemeijer et al [35]       | Private  | 50     | 1.000 | 0.870 |       |       |
| Quellec et al [20]         | Private  | 1739   |       |       | 0.927 |       |
| Saleh et al [153]          | Private  | 98     | 0.843 | 0.936 |       |       |

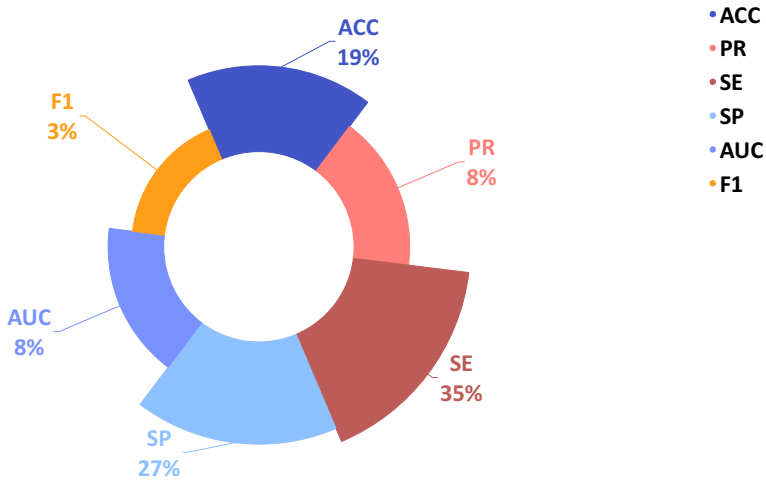
sensitivity (True Positive Rate (TPR)), recall, precision, accuracy, F1 score, Area Under the receiver operating characteristic Curve (AUC), Free-Response ROC (FROC).

The primary goal of these metrics is to evaluate the performance based on True Negatives (TN), True Positives (TP), False Negatives (FN), and False Positives (FP). The TP represents the count of pixels, MAs or images that the algorithm correctly identifies as positive cases, which aligns with the assessments made by expert clinicians. The FN refers to instances where the algorithm incorrectly classifies a case as negative, failing to detect an actual label. Conversely, the TN indicate the algorithm's correct identification of non-MA elements. FP occurs when the algorithm wrongly identifies something as an MA, which human experts would not consider as such.

Benchmarking can help us to evaluate the effectiveness of early DR detection. It checks if these systems can be trusted to make the right predictions when it's really important for safety-critical scenarios. It offers a series of practical tasks that truly represent the challenges found in real-world situations. These tasks need accurate methods to predict and measure uncertainties. The benchmark evaluates traditional and state-of-the-art Bayesian deep learning techniques, measuring their performance using evaluation metrics for each task [154].

Further, an algorithm is proposed for MA detection in fundus images. The paper used benchmarking to evaluate the effectiveness of early DR detection. They used the Area Under the Precision-Recall curve (AUPR) as a key performance metric [155]. Moreover, establishing and sharing benchmark databases with confirmed accurate data and powerful evaluation methods is essential for improving algorithms and comparing different methods. The DiaRetDB1 V2.1 database is a good example of DR detection algorithms. It includes retinal images, expert-verified ground truth, and a basic lesion detection algorithm [102].

Table 5 shows how different researchers results on public datasets for MA detection. It covers their ACC, SE, SP, AUC, and F1 score. Moreover, Table 6 shows the performance results of researchers using private datasets to detect MAs. It includes their results for SE, SP, AUC, and F1 score. Finally, Fig. 7 presents the distribution of performance measure metrics in the research studied. It gives percentages for each performance measure metrics: ACC is used by 19% of the studies, PR 8%, SE 35%, SP 27%, AUC 8% and F1 score 3%, respectively.



**Fig. 7** The distribution of performance measure metrics in the research studied

## 4 Microaneurysm detection

The initial research used fluorescence angiography, which offered great MA contrast. However, recent methods focus on detecting MAs in retinal photos. Consequently, fundus images have emerged as the primary imaging choice in screening for diabetic retinopathy due to their cost-effectiveness, efficiency, and simplicity [10].

Detecting MAs plays a crucial role in diagnosing DR. Various methods have been proposed to simplify MA detection in retinal images, leading to numerous research papers and comprehensive reviews [11–13]. For example, Biyani and Patre [13] classified present methods into the following five groups: (a) methods based on mathematical morphology and region growing, (b) supervised classification methods, (c) methods based on pattern matching and vocabulary learning, (d) deep learning (DL) based methods and (e) some other methods.

### 4.1 Image processing methods

A comprehensive analysis of 23 studies has been conducted, focusing on image processing techniques for detecting MAs in retinal images, [9, 10, 14–34]. Prior research has introduced numerous automatic detection approaches for MAs in color fundus images. These methods can be broadly classified into several categories, including morphological-based techniques, template matching, supervised learning or a combination of both [15–18].

#### 4.1.1 Morphological techniques

Morphological-based methods have been employed to improve the accuracy of MA detection, as reported in Fleming et al. [14] and Tavakoli et al. [19]. While these methods are typically fast and straightforward, their simplicity can sometimes lead to suboptimal detection, a limitation highlighted by Wang et al [18]. Morphological operations such as closing [17] and top-hat transformation are particularly chosen to target MAs, given their characteristically uniform and small shapes. Wang et al. [18] introduced an object filtering technique to isolate

potential MAs. The profiles of these potential objects were then analyzed using singular spectrum analysis. Finally, A k-Nearest Neighbors (kNN) classifier, alongside statistical profile features, is then used to differentiate between MA and non-MA.

However, despite these advancements, it is important to note that the effectiveness of mathematical morphological approaches largely depends on the appropriate selection of structural elements. Regardless, in addition to the above problems, most mathematical morphological approaches rely on correctly selecting structural elements. These methods may become less effective if there is a need to adapt the shapes and sizes of the structural elements to encompass the variability of MAs, as discussed by Tavakoli et al [19].

#### 4.1.2 Template matching

Template matching approach for MA detection are inspired by the characteristic that an MA resembles a Gaussian peak across various projections, and its intensity distribution is consistent with a Gaussian distribution, as noted in Quellec et al. [20] and Zhang et al. [21].

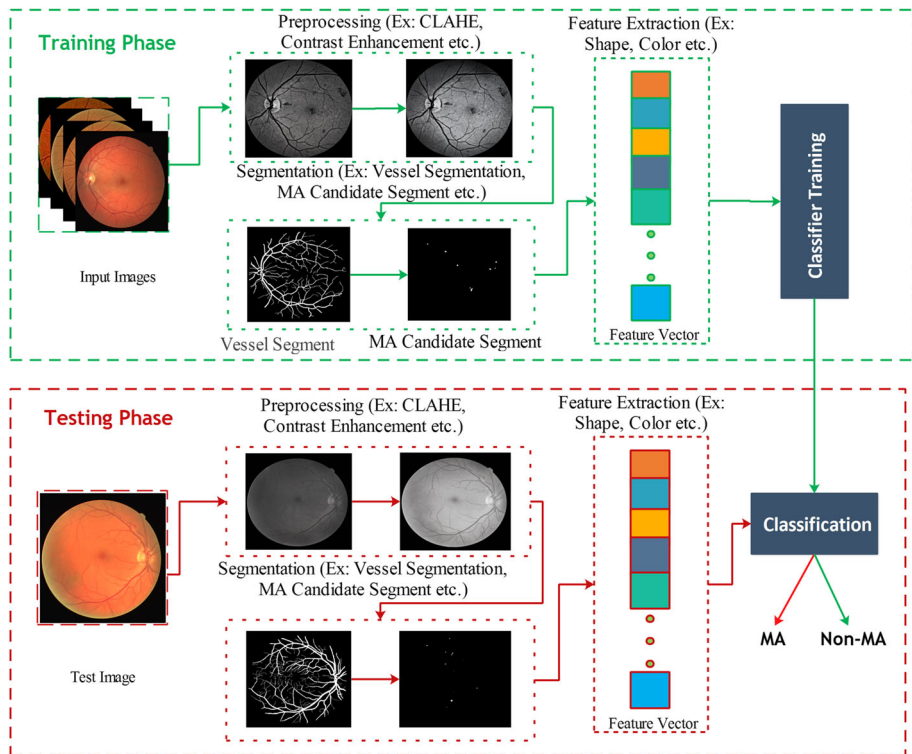
Quellec et al. [20] utilized a wavelet transformation for the detection of MA. Zhang et al. [21] applied a multi-scale correlation coefficient technique combined with nonlinear filtering, employing five Gaussian kernels with varying standard deviations to identify MA candidates. Meanwhile, Zhou et al. [22] proposed an unsupervised learning approach for MA detection. Instead, their system uses a novel unified statistic to differentiate actual MAs from non-MAs. However, the algorithm is limited by its inability to detect subtle and hidden structures, similar to morphological methods, and it also requires the determination of multiple parameters based on empirical evidence.

Dashtbozorg et al. [9] provided an innovative technique for the accurate and reliable detection of MAs in retinal images. In the first step, potential MA candidates are identified using gradient weighting and iterative thresholding methods. Subsequently, new features based on local convergence index filters, as well as shape and intensity descriptors, are extracted for each candidate. A combination of these features is then provided to a hybrid ensemble of classifiers, including a boosting classifier, to differentiate between MAs and non-MAs. Additionally, the study in [23] proposes a novel method for detecting HEs and MAs in color fundus photographs. This technique involves a set of dynamic shape features tested across 6 different datasets.

#### 4.1.3 Supervised learning

Machine learning methods like supervised classification detect MA in retinal images. Most methods have a supervised classification step after candidate extraction because candidate classification has been shown to play an essential role in minimizing FPs. Different approaches have been developed due to the extensive range of variables that might be calculated in MA candidates and the wide range of classifiers available. For example, Niemeyer et al. [35] offered a new technique that adds a new feature set (color-based, gaussian filter output, and shape based) to the features provided by Frame et al. [36]. The authors examined the performance of the linear unique, quadratic discriminative, and kNN classifiers. They obtained the best results utilizing the kNN classifier. The method presented by Mizutani et al. [37] used a double-ring filter to search for initial MA candidates and removed the blood vessels. Then, 12 shape and intensity features were extracted and utilized with a rule-based method and artificial neural networks.

The MA detection method is shown here in detail and can be found in Fig. 8.



**Fig. 8** Microaneurysms detection using image processing based method

The figure outlines an integrated machine-learning pipeline for detecting MAs in retinal images, a critical marker for diagnosing retinal diseases such as diabetic retinopathy. The process begins with a training phase where a collection of retinal images is acted as the input. These images are then subjected to preprocessing techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE), to enhance contrast and facilitate the distinction of retinal features. Following preprocessing, segmentation is carried out to isolate the retinal blood vessels and identify potential MA candidates within the images. The next critical step is feature extraction, where characteristics such as shape and colour are derived from the segmented candidates, creating a descriptive feature vector for each potential MA. These vectors encapsulate the essential attributes necessary for the final stage of classifier training. After the classification process, the system conclusively presents the output, indicating the presence of an MA or non-MA, thus automating the detection of MAs with precision and benefiting in timely eye diagnosis. Moreover, Table 7 shows the image processing-based methods of MA detection using retinal fundus images.

## 4.2 Deep learning methods

DL is a subset of artificial intelligence that relies on Artificial Neural Networks (ANN) inspired by the human brain's learning process. DL automatically uncovers intricate relationships within input data using mathematical techniques. Unlike traditional supervised

**Table 7** Review of image processing based techniques for MA detection

| Ref.                  | year | Method                                                                                                                  | Features                                          | Database(s)                              | Performance                               |
|-----------------------|------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------|-------------------------------------------|
| Wu et al [27]         | 2017 | kNN classifier to differentiate proper MAs from spurious candidates                                                     | Local features and Profile features               | ROC, e-optha                             | ----                                      |
| Bharkad [28]          | 2017 | The blood vessels were removed using a grayscale morphological top-hat transform to detect MAs.                         | shape and size-based features                     | DiaretDB1                                | ----                                      |
| Ren et al [29]        | 2017 | A robust over-sampling approach is used with ensemble classifiers (boost, bagging, and random subspace).                | intensity, shape, and gradient distribution,      | DiaretDB1, ROC                           | SE = 0.961, AUC = 0.962                   |
| Dashtbozorg et al [9] | 2018 | MA candidate extraction techniques using a set of new features.                                                         | local convergence index filters                   | e-optha, DiaretDB1, RG-RGB-MA, RC-SLO-MA | AUC = 0.798, SE = 0.820                   |
| Derwin et al [30]     | 2020 | ANN classifier                                                                                                          | Local neighborhood differential coherence pattern | ROC                                      | FRCC score = 0.481                        |
| Melo et al [10]       | 2020 | A RUS-Boost classifier is utilized to classify MA and non-MA objects.                                                   | Contrast, Intensity, and shape-based features     | ROC, e-optha                             | AUC = 0.83, SE = 0.68, 0.90               |
| Mazlan et al [31]     | 2020 | Multilevel thresholding and MLP classification                                                                          | Shape, statistical, color                         | e-optha                                  | ACC = 92.88%                              |
| Tavakoli et al [34]   | 2021 | MA detection and numbering using RT and supervised SVM classifier.                                                      | Not Mentioned                                     | DiaretDB1                                | SE = 0.957, SP = 0.954, AUC = 0.97        |
| Zhang et al [38]      | 2022 | Local fourier transforms and analyzing them to extract critical features, and identifying potential points of interest. | statistical and shape                             | ROC, e-optha                             | SE = 0.89, 0.73 AUC = 0.961               |
| Soares et al [39]     | 2023 | The algorithm utilizes a multi-scale framework to effectively detect MA.                                                | multi-scale features                              | ROC, e-optha                             | SE = 0.936, AUC = 0.874                   |
| Zhang et al [40]      | 2023 | Hemorrhages and MAs detection with shape features in color fundus images                                                | Shape, size, intensity and statistics             | e-optha                                  | SE = 0.89, FI score = 0.507, AUC = 0.9821 |
| Perumal et al [41]    | 2023 | MA detection via local fourier transform and neighborhood analysis.                                                     | multi-scale features                              | MESSIDOR                                 | ACC = 0.9931, SE = 0.9811, SP = 0.9961    |

learning methods, DL requires less human intervention and doesn't rely on predefined features. This makes DL particularly effective for analyzing complex medical images, as its components can learn from detailed data automatically

The characteristics of retina imaging play a vital role in early DR diagnosis. In the below section, In the upcoming section, we'll explore findings from 22 research papers [16, 33, 42–48, 51–63] that revealed various deep learning-based methods for early DR diagnosis through automated MA detection. This discussion covers methods, feature extraction techniques, databases used, classification approaches, and performance metrics, outlined in Table 8.

The presence of MAs in fundus images serves as an early sign of DR. Implementing an automatic computer-based classification system can enhance the screening efficiency for a vast number of retinal images. However, detecting MAs is challenging due to their small size, pathology, and variations in anatomy, leading to noise, low contrast, and irregular illumination in images. Utilizing computerized diagnostic processes can aid in large-scale fundus image screening, accessible even to untrained healthcare professionals. Recent advancements in deep learning have facilitated rapid and automated DR detection, displaying promising outcomes, particularly at the pixel level [64] or patch-level supervision [65–67].

Several effective methods are used for detecting MAs. Zhang et al. [33] introduced a method using original fundus image channels and additional samples with improved contrast and concealed backgrounds. Liao et al. [51] developed an encoder-decoder network emphasizing contrast between network layers. Their custom loss function helped focus on complex examples during training. Xia et al. [52] presented a two-stage network, using multiple input scales and filtering false positives for efficient feature extraction in MA detection.

Adal et al. [55] have introduced another machine learning-based method for detecting MAs. Firstly, the normalized fundus images using preprocessing methods and features were extracted using multiple scale descriptors. Finally, the semi-supervised approach was used to train a model to classify MA and non-MA pixels with 0.36 AUC. Moreover, different techniques cannot overcome the problem of overfitting large-scale datasets and having a low discriminative power scale. In identifying MAs and DR, Li et al. [56] developed an application for automatic diagnostics based on machine learning.

Moreover, the author describes a Convolutional Neural Network (CNN) DR detection and classification system. Similarly, Suriyal et al. [57] presented a real-time mobile assistance app based on a deep learning technique for diabetic retinopathy detection. It uses a deep neural network model with TensorFlow to train and analyze fundus images. Further, Jadhav et al. [58] proposed early detection of the MAs method based on the Gabor filter and acquired results of 76.75%, 78.76%, and 80.85% in sensitivity, specificity, and accuracy, respectively, using an Indian dataset.

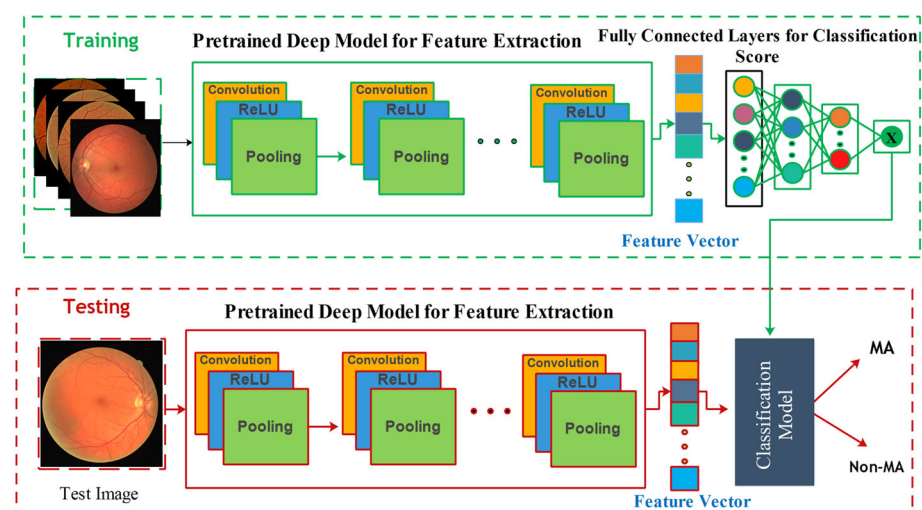
Furthermore, another study presented a Region-CNN (R-CNN) based microaneurysm detection method. Using the described method, features are extracted and classified using VGG-16 to classify candidate regions [68]. MAs were detected using a different technique that used statistical and stock well transform features to distinguish between normal and diseased images. During the training and testing phases, 1140 retinal images were used in total [59]. Mateen et al. [60] presented a new method based on a DL approach for early MA detection. The proposed method uses pre-trained CNN VGG-19 and Inception-v3 models with a feature embedding strategy. The model was tested on E-Ophtha and DIARETDB1 publicly available datasets and got classification accuracy of 96% and 94%, respectively. The proposed [61] system is trained using a deep CNN using a reinforcement learning technique. Further, different patches of the color image are acquired from ground-truth samples of MA and non-MA locations to train the DCNN. The proposed approach used the ROC training dataset for validation.

**Table 8** Review of Neural Network and Deep Learning based techniques used to detect MAs

| Ref.                 | year | Method                                                                                                                                      | Preprocessing                                                                                                                                  | Database(s)   | Model, Layers, Activation 'or' Loss function     | Performance                                                                                                                              |
|----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Dai et al [42]       | 2017 | A multi-sieving CNN (MS-CNNs).                                                                                                              | Multi-Sieving Convolutional Neural Network (MS-CNN)                                                                                            | DIARETDB1     | - - - - -                                        | SE = 0.878, PR = 0.997, ACC = 0.961                                                                                                      |
| Orlando et al [43]   | 2018 | Propose combining deep learning and hand-crafted features to develop discriminative models for red lesion identification.                   | Hybrid feature vector.                                                                                                                         | DIARETDB1     | - - - - -                                        | AUC = 0.8932, 0.9347 Sen = 0.9109, (CPM) = 0.4874, 0.3683 0.9721 and Competition Metric                                                  |
| Chudzik et al [44]   | 2018 | Proposed a patch-based fully CNN using batch normalization layers. A new clinical report a guided method to detect MAs from Retinal images. | Green plane, morpho-logical operations, Otsu thresholding Pre-trained Alex Net, random feature space partitioning, text information extraction | e-Ophtha.     | Deep U-Net                                       | sensitives at various FPIs ±0.116                                                                                                        |
| Dai et al [16]       | 2018 |                                                                                                                                             |                                                                                                                                                | DIARETDB1 ROC | network (14, Sigmoid, Dice coefficient function) | (ROC), 0.392 ±0.157 (DB1), 0.193 ±0.233 (e-Ophtha)                                                                                       |
| Eftekhari et al [45] | 2019 | Introduce a novel deep CNN model.                                                                                                           | Color normalization, background E-Ophtha.                                                                                                      | DIARETDB1     | MS-CNN, Image-to-text mapping, FCNs              | SE = 0.878, PR = 0.997, ACC = 0.961, F1 Score = 0.934                                                                                    |
| Kou et al [46]       | 2019 | Using the deep residual and recurrent convolutional procedures, a deep recurrent U-Net (DRU-Net) was created.                               | subtraction and image patches deep recurrent U- created. Net (DRU-Net) was Image patches                                                       | ROC           | CNN (13, softmax, CEL)                           | Final score CPM=0.461, FAUC = 0.660 including Average sensitivity at seven false positive rates. ACC = 0.999, 0.999 , AUC = 0.994, 0.982 |

Table 8 continued

| Ref.                | year | Method                                                                                                 | Preprocessing                                                              | Database(s)              | Model, Layers, Activation 'or' Loss function | Performance                                            |
|---------------------|------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------|----------------------------------------------|--------------------------------------------------------|
| Guo et al [47]      | 2019 | Presented a novel L-Seg framework, a unique method for segmenting fundus images with multiple lesions. | image scaling to $1440 \times 960$                                         | DDR, IDRiD, E-Ophtha     | L-Seg (5, -, multi-channel loss)             | mAUC = 0.6515 (IDRiD), 0.3208 (DDR), 0.2929 (e-Ophtha) |
| Zhang et al [33]    | 2021 | Proposes an efficient approach based on feature-transfer learning and background.                      | contrast enhancement, Green and blue channel                               | ROC, DIARETDB1, E-Ophtha | Sparse auto-encoder (1, Sigmoid, CEL)        | CPM = 67.4 (ROC), 99.5 (DB 1), 94.9 (E-Ophtha)         |
| Hervella et al [48] | 2022 | Introduce a new deep learning method with unlabeled multimodal image pairs.                            | self-super-vised fashion                                                   | E-Ophtha, ROC, DDR       | - - - - -                                    | AUROC (%) = 97.46 $\pm$ 0.69 (E-Ophtha)                |
| Jian et al [49]     | 2023 | Triple-DRNet model that integrates two efficient attention modules in subnetworks.                     | Gaussian blur is applied to the image, subtracting the local average color | APTOS 2019               | PDR-Net module                               | ACC = 0.9208                                           |
| Liu et al [50]      | 2023 | Developed a classification model, named Triple-DRNet, for grading DR.                                  | Gaussian filter, HE, and CLAHE to enhance contrast and brightness          | DeepDRiD                 | ResNet50                                     | F1 score = 0.6559                                      |



**Fig. 9** The process for detecting MA and Non-MA using a deep learning technique

Figure 9 illustrates the process for detecting MA or Non-MA utilizing a deep learning technique. The system uses a medical image processed through multiple convolutional layers to extract features in the training phase. These features are then passed through ReLU layers to add non-linearity and pooling layers to reduce dimensionality, resulting in a condensed feature vector. This vector is fed into fully connected layers that classify the image and assign a score indicating the presence of certain conditions. A new image undergoes the same feature extraction process in the testing phase. The extracted feature vector is input into a pre-trained classification model, which predicts whether the image is indicative of MA or Non-MA.

## 5 Microaneurysm segmentation

Medical imaging segmentation is a significant area of research that involves separating images into meaningful structures and finding areas of interest. Medical image segmentation can be performed in different aspects, including image processing, ML and DL. In this answer, we will discuss the main differences between these two ways of doing things.

Image processing-based segmentation includes a series of algorithms and mathematical operations on an image to improve its quality and pull out the important parts [69]. The image processing-based method uses what you currently know about the image and specific image processing techniques like edge detection, thresholding, and morphological operations to separate the different parts of an image.

DL based medical imaging segmentation is a type of machine learning that uses deep neural networks to automatically segment images. DL models can learn from a large set of labelled images and find complex patterns in the data, making them good for segmenting medical images. CNN and Multi-Task ANN (MTANN) are two common deep learning architectures used in medical image segmentation [69, 70].

On the other hand, machine learning-based medical imaging segmentation uses a set of images and the labelled data that goes with them to train a model that can automatically segment new images. This method needs a large set of labelled images and a learning algorithm,

like supervised or unsupervised learning, to find patterns in the data and make a segmentation model [70].

MA segmentation can be a tricky and challenging task due to many reasons. Firstly, MAs are much smaller than other parts of the retina, making them hard to spot [71]. Also, the low contrast of the lesion and things like noise and the difference between the lesions and their background can make segmentation harder [72]. The complex backgrounds in fundus images and different lighting conditions make automatic detection challenging [73]. Furthermore, another big issue is the bad quality of MA datasets, creating significant image analysis problems [39]. Lastly, these datasets often have uneven positive and negative samples, requiring special methods [74].

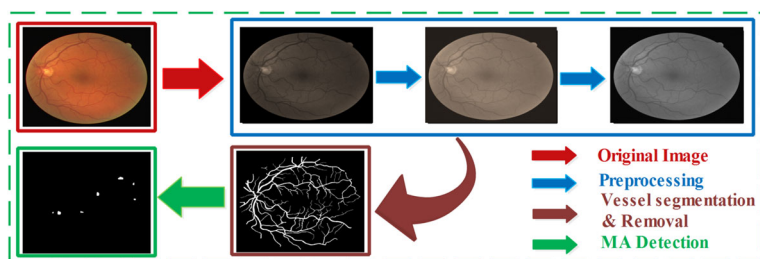
Segmenting MA is significant in diagnosing and treating DR. It helps doctors determine if someone has the disease, how bad it is, and what the stage of the disease is [75]. MAs are one of the first signs of DR, so finding them is key for checking if someone has the condition [76]. MA arise as small red isolated spots next to small blood vessels and should occur in the early DR process. Segmenting these tiny spots helps doctors figure out where they are and keeps track of how the disease changes over time [75]. It's crucial to find and treat diabetic retinopathy in the early stage; otherwise, it could lead to vision loss [77]. Some new methods like C2AL and MSNet work well in finding and marking these microaneurysms, making it easier for doctors to diagnose and help them treat diabetic retinopathy [52].

## 5.1 Image processing methods

Since MAs are the initial clinical symptom of DR, they have gained the attention of researchers. It is still challenging to perform automatic MA segmentation because of the small size of MAs and the low local contrast of the image. Many constraints affect MA segmentation, including clarity, background texture, other lesions with similar colors, and variations in image illumination. The segmentation of MAs is significant for early diagnosis and treatment. Therefore, two successful methods are often used for MA segmentation: image processing-based and Machine learning based. Recently one of the most common approaches to the medical image segmentation task is a deep learning-based algorithm such as U-Net. Mazlan et al. [31] developed a simplified technique based on morphological transforms for MAs Detection by removing retinal capillaries and other undesirable lesions.

Meanwhile, [78] agreed with this strategy and used a series of morphological operators that had their settings optimized for optimal performance in order to identify the MAs. This procedure proved difficult because it was examined using retinal photographs taken with the pupil non-dilated and with low contrast. Sopharak et al. [78] improved on previous studies by employing a simple hybrid morphological-based technique. In the segmentation stage, the naive bayes classifier was used to enhance the efficiency of the results. Zubair et al. [79] put dynamic thresholding into place by picking the threshold value for each image after figuring out its range of brightness, and further pixel values were used to calculate this range. Since the threshold value depends on intensity-based characteristics, the image set collected under varied lighting conditions had greater accuracy, 99.72%, sensitivity, 98.73%, and specificity, 98.25%. Long et al. [80] used fuzzy c-means clustering to determine the local dynamic threshold of each 30 x 40 sub-image. When there was a low contrast between hard exudates and the background, it was revealed that threshold-based techniques gave a low performance.

Figure 10 likely illustrates segmenting MAs from retinal fundus images using various image processing techniques. The segmentation process typically involves the following



**Fig. 10** MA segmentation utilizing image processing technique

steps. In the first step, preprocessing may include enhancing the image quality, removing noise, and normalizing contrasts to make MA more detectable. In the second step, feature extraction, key features that help identify MAs are extracted. These could include shape, size, colour, texture, or other relevant characteristics distinct to MAs. In the third step, the actual Segmentation of MAs occurs. The algorithm will apply specific rules or machine learning models to differentiate between MAs and non-MAs areas. In the fourth step, post-processing, the segmented image may undergo further processing to refine the results, remove false positives, and improve accuracy. Finally, the final output is a segmented image where MAs are identified and marked, which is crucial for further MA analysis and diagnosis.

## 5.2 Deep learning methods

The drawbacks of the above methodologies are solved by machine learning techniques, particularly deep learning. Furthermore, supervised and unsupervised techniques have been the research focus recently. MA segmentation is vital for diagnosing DR. Many deep-learning approaches have been developed specifically for this purpose. The main goal of these approaches is to improve the accuracy of segmentation.

The study presented in paper [47] introduces a novel L-Seg network designed to segment small image lesions. It addresses the challenge of small lesion regions needing to be detected effectively at high levels of the network by using a method that fuses features from multiple scales. Besides, the paper presents a new multi-channel bin loss approach to tackle class and loss imbalances in segmentation tasks. L-Seg's performance was evaluated on three fundus image datasets, including two publicly available ones, IDRiD and e-optha, and a private dataset DDR. Moreover, Liu et al. [93] used the M2MRF method for MA segmentation. This method simplifies the process by merging features from a large area into a smaller, more manageable space, resulting in several concise outputs. This technique keeps details about small lesions and captures long-term patterns to improve lesion identification. Furthermore, the method was tested on two lesion segmentation benchmarks (DDR and IDRiD) and showed that M2MRF performs better than other current methods.

Garifullin et al. [94] introduced a new system with thorough testing to analyze how well it segments lesions, calibrates models, and predicts with uncertainties. It overcomes issues in deep probabilistic modelling and measuring uncertainty. On the IDRiD dataset, their method reached a 0.484 score in the precision-recall area, a 0.434 sensitivity rate, and a 0.999 specificity rate for detecting MAs. Building on advancing DR lesion detection, another method,

EAD-Net [81] was proposed to identify different lesions in DR using CNN. This network has three parts: an encoder, an attention module, and a decoder. We first prepare eye images and then use EAD-Net to find features and predict lesions. The method was very accurate, with a sensitivity of 92.77%, a specificity of 99.98%, and an accuracy of 99.97% on one dataset, and it performed well on another dataset. Further complementing these advancements, a third study [95] developed a specialized U-Net model using the pre-trained Inception-V3 CNN network, employing a unique shuffling and sub-pixel convolution technique. This model was trained and tested on two major datasets, IDRiD and DIARETDB1, for detecting various diabetic retinopathy conditions like MAs and HEs. The IDRiD dataset achieved an impressive 99.95% accuracy, 70.76% sensitivity, and 99.98% specificity for MA segmentation.

The study [96] introduces a modified UNet model that combines a residual network with periodic shuffling and sub-pixel convolution. This new architecture was trained to identify MA and EX using IDRiD and e-optha datasets. The IDRiD dataset achieved 99.88% accuracy, 99.85% sensitivity, and 99.95% specificity with a high dice score for segmenting MA and EX. Further, training on e-optha and testing on IDRiD resulted in 99.98% accuracy and similarly high sensitivity, specificity, and dice scores for MA detection. The paper [97] introduced a new network named CLC-Net that combines contextual and detailed patch information and an attention mechanism to segment various DR lesions simultaneously. The method's multi-task learning strategy enhances the segmentation's accuracy and robustness. It was further tested on three different fundus segmentation datasets, showing that it performs exceptionally well compared to existing techniques.

Further expanding the scope of research in this field, paper [98] introduces a new algorithm called Poisson Blending Data Augmentation (PBDA) to create synthetic images for expanding training data in lesion segmentation. Through extensive testing, it's found that position constraints and synthesis density of one lesion type affect the segmentation of others. Additionally, a new neural network architecture, DSR-U-Net ++, is proposed for segmenting multiple types of DR lesions simultaneously. Upadhyay et al. [99] proposed a red lesion segmentation algorithm that has the potential to screen out the cases of DR in the early stages by combining five lesion-specific, manually created features with U-net deep features. Furthermore, the paper introduces a new 'characteristic patch-based training' for more effective red lesion targeting. The study shows these additions improve the baseline model and method, surpassing others on IDRiD and DDR datasets by about 7% and 10% in the precision-recall curve area. Moreover, Pappu et al. [82] used MRT-SegNet with special encoder blocks. These blocks use MRT filters to get texture features from retinal images and combine them with local features from the network's usual encoder. This combination helps the network to identify very small lesions on the retina. We tested this model on the IDRiD dataset. Furthermore [83] introduces an algorithm for detecting red lesions in eye images, focusing on HEs and MAs. The algorithm's effectiveness was tested on several datasets, including ROC challenge and Messidor, using metrics like accuracy and precision.

In a series of studies [86][87][88][89][100], various innovative approaches have been developed for improving the segmentation and detection of lesions in medical images, particularly focusing on DR and its associated features in retinal images. The paper [86] presents TC-Net, a straightforward yet effective framework for medical image segmentation. It combines both local and broad-range details from these images. The method uses a CNN for encoding and decoding, focusing on local image features. TC-Net scored 0.6985 on the IDRiD database and 0.5171 on the DDR database. Pavani et al. proposed [87] a new deep-learning RILBP-YNet network that introduced a new way to identify different retinal lesions

at the same time. This network uses Local Binary Patterns to get textural features, which helps improve the process of identifying these lesions. This model is not just good at finding DR lesions but also accurately identifies the optic disc (a part of the eye). It has an AUC score of 0.7108, which is a way to measure how well it performs this task.

In another innovative approach, study [88] introduced Trans2U-Net, an innovative multi-view technique designed for the effective detection of various retinal lesions, showcasing a novel approach in the field. This system uses a two-level network, merging the local-detail capabilities of U-Net with the global attention mechanism of TransU-Net, offering a comprehensive approach to lesion. Acharya et al. [89] employed a one-dimensional segmentation process based on the u-net model for detecting MAs. This method involves extracting 1-D MA features from preprocessed eye images using an optimized line drawing algorithm. We tested its effectiveness by measuring sensitivity against different false positive rates in the e-optha and ROC datasets. Further, study [100] introduces a new automatic method for detecting MAs in color fundus images, involving breaking down images into patches, using segmentation models for analysis, and then reconstructing the segmentation map. The method combines three deep networks (U-Net, ResNet34-UNet, and UNet++) and shows high effectiveness, achieving a Dice score of 0.95 and an IoU of 0.91, indicating its strong potential for early DR detection.

The paper [101] describes a multistage deep learning method for segmenting MAs using a multi-scale patch-wise approach. This work's main contributions are embedding triplet loss with selective sampling and fusing segmentation on multiple scales for MA detection. The research study by [96] introduces a U-Net architecture, a type of neural network, improved with techniques like periodic shuffling and sub-pixel convolution. This network was trained and tested for identifying MAs using the IDRiD and e-optha datasets. It achieved very high accuracy of 99.88%, sensitivity of 99.85%, and specificity of 99.95%, along with a 0.9998 dice score, in detecting MAs in the IDRiD dataset. This was part of their work on lesion segmentation.. Wen et al. [81] suggested a convolutional neural network architecture called EAD-Net. Encoder, dual attention, and decoder modules are the three categories that can be used to describe the architecture. The EAD-Net accomplished a high performance compared to other advanced methods on publicly available datasets. The EDA-Net achieved a sensitivity of 92.77%, a specificity of 99.98%, and an accuracy of 99.97%.

This paper [92] presents a method for segmenting four common DR lesions using CNNs. First, the colour fundus images are preprocessed and resized. Then, a group of Fully Convolutional Networks (FCNs), each taking inputs of different sizes, are trained to identify lesions at various scales. An additional CNN is used to combine and improve the results from these FCNs. Collectively, these studies represent significant advancements in medical image analysis, particularly in the context of DR. Each approach, whether focusing on innovative network architectures or specialized segmentation techniques, enhances early detection and accurate diagnosis in the medical imaging field.

Current research concludes that deep learning can improve the development of an effective unified segmentation system for detecting and grading retinal lesions. In addition, the added benefit of automatic feature extraction allows for more precise retinal boundary localization than earlier methods. Because of their robustness and flexibility, DL-based systems will be a significant research study nowadays. Table 9 in the document is a clear and well-organized list that shows different deep learning-based segmentation methods for detecting MAs. This comprehensive table thus offers a critical overview of the current state of deep learning techniques in MA detection.

**Table 9** Review of deep learning based segmentation techniques for detecting MAs

| Ref.              | Year | Method                                                                                                                        | Class            | Dataset(s)                              | Performance                                                           |
|-------------------|------|-------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|-----------------------------------------------------------------------|
| Wan et al [81]    | 2021 | EAD-Net network to identify different lesions in diabetic retinopathy using a CNN.                                            | multi            | e-optha,<br>IDRiD                       | SP = 0.9998,<br>ACC = 0.9991,<br>F1 score = 0.2682                    |
| Pappu et al [82]  | 2022 | MRT-SegNet is for the segmentation of retinal lesions and OD simultaneously.                                                  | multi            | IDRiD                                   | ACC = 0.998, SP = 0.999, Sen = 0.397, F1 score = 0.792, PR 0.715      |
| Latha et al [83]  | 2022 | Hexagonal pattern-based red lesion detection with two-level segmentation in fundus images                                     | multi            | ROC, e-optha,<br>DiaretDB1,<br>Messidor | ACC = 0.9578, SE = 0.8455, PR = 0.9731, F1 score 0.9047, SP = 0.8681  |
| Kundu et al [84]  | 2022 | This study concentrates on segmenting red lesions utilizing a nested U-Net architecture                                       | single           | DIARETDB1                               | ACC = 0.9953, SP = 0.996, SE = 0.887, F1 score = 0.7921, PR = 0.7150  |
| Guo and Peng [85] | 2022 | DARNet, an enhanced RefineNet model, is developed for the automated segmentation of multiple lesions in diabetic retinopathy. | multi            | IDRiD,<br>e-optha,<br>DDR               | ACC = 0.9582,<br>0.9617, 0.9578,<br>SP = 0.9998                       |
| Zhang et al [86]  | 2023 | A novel TC-Net framework                                                                                                      | multi            | IDRiD,<br>DDR                           | AUC = 0.9881,<br>0.9758                                               |
| Pavani et al [87] | 2023 | A novel RILBP-YNet network that provided a novel path for segmenting simultaneous retinal lesions                             | multi<br>e-optha | IDRiD,                                  | SE = 0.280, SP = 0.999, ACC = 0.9987, F1 score = 0.3602, AUC = 0.7108 |
| Zhang et al [88]  | 2023 | Trans2U-Net system multi-view approach by merging U-Net's local focus with TransU-Net's global attention.                     | multi            | IDRiD,<br>DDR,<br>e-optha               | Mean ROC = 0.9819,<br>0.8995, 0.9614                                  |

Table 9 continued

| Ref.               | Year | Method                                                                                                                        | Class | Dataset(s)                      | Performance                                  |
|--------------------|------|-------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------|----------------------------------------------|
| Acharya et al [89] | 2023 | 1-D U-Net Based Microaneurysm Detection through Feature Sequence Segmentation                                                 | multi | e-ophtha, ROC                   | CPM = 0.592, 0.802                           |
| Saranya et al [90] | 2023 | Automated UNet-based system for early detection of diabetic retinopathy by segmenting red lesions.                            | multi | IDRiD, DIARETDB1, MESSIDOR      | acc = 0.956, SP = 0.999, SE = 0.890          |
| Ullah et al [91]   | 2023 | Semi supervised multi-task decoders network (SSMD-UNet) for segmenting diabetic retinopathy                                   | multi | FGADR, IDRiD                    | ACC = 0.926, 0.901                           |
| Guo et al [92]     | 2024 | CNN-based method for segmenting diabetic retinopathy lesions, using resized CFP images and multiple FCNs for scale variation. | multi | IDRiD, DDR, e-ophtha, DIARETDB1 | AUPR% for MA = 0.4975, 0.194, 0.3038, 0.4677 |

## 6 Discussion

The automated detection and segmentation of MAs are crucial for early diagnosis and monitoring of DR. In this review paper, we showed that with advancements in ML and DL, automated methods have shown significant promise. Here are some recent trends and potential future directions in this field.

### 6.1 Recent trends

CNNs have become the state-of-the-art for image classification, detection, and segmentation tasks, including identifying MAs. More sophisticated architectures like U-Net for segmentation and specialized object detection networks like Faster R-CNN and YOLO (You Only Look Once) [156] have been adapted for detecting small lesions such as the MAs in this paper. Given the limited size of annotated medical image datasets, transfer learning, where a model pre-trained on a large dataset is fine-tuned on medical images, is increasingly used to improve performance in MA detection. Techniques to artificially increase the size and variability of training datasets, such as geometric transformations and intensity modifications, help improve the robustness and generalization of models for MA detection.

As deep learning models are often seen as "black boxes," there is a growing emphasis on making deep learning models more interpretable, especially in the medical field, where understanding the reasoning behind a model's decision is crucial. Techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) [157] provide visual explanations for model decisions, which is critical in medical applications for gaining the trust of clinicians.

To address privacy concerns and data-sharing issues, federated learning allows models to be trained across multiple decentralized devices or servers holding local data samples without exchanging them, which is particularly useful for medical data.

### 6.2 Potential future directions

Future directions in this field will likely involve further development of these trends, emphasising improving the accuracy and robustness of segmentation models, ensuring their explainability and interpretability, and facilitating their deployment in real-world clinical settings.

- **Ensemble Methods:** Combining predictions from multiple models or using ensemble learning techniques can often lead to more accurate and robust systems for MA detection.
- **Attention Mechanisms:** Integrating attention mechanisms into deep learning models can help the network focus on relevant regions of the image. This is particularly useful for identifying small and subtle lesions like MAs.
- **Few-shot Learning and Zero-shot Learning:** These approaches aim to develop models that can detect MAs with very few or even no labeled examples. This is particularly important for rare conditions or when annotated medical data is scarce.
- **Multi-Class Segmentation:** Recent methods focus on not just detecting MAs but also other lesions such as EXs, HEs and SEs. Multi-class segmentation models are designed to differentiate between various lesion types in retinal images.
- **Multimodal Imaging:** Using data from different imaging modalities (such as color fundus photography, optical coherence tomography and fluorescein angiography) in an integrated manner may provide a more comprehensive understanding of the retinal con-

dition and improve the detection of MAs. Advances in retinal imaging technology, such as adaptive optics and ultra-widefield imaging, can provide higher-resolution images and a wider field of view, aiding in the detection and segmentation of MAs.

- **Real-time Detection:** With the improvement in computational efficiency, there is a potential for developing real-time MA detection systems that can be used during retinal examinations.
- **Clinical Trials and Validation Studies:** More extensive clinical trials and validation studies are needed to assess the real-world performance and reliability of automated MA detection systems.
- **Dataset and Model Standardization:** Efforts are being made to standardize datasets and model architectures to facilitate comparison and reproducibility of results. Publicly available benchmarks and challenges also help in pushing the field forward.

Overall, this field is moving towards more sophisticated, reliable, and interpretable systems for MA detection and segmentation, emphasizing clinical applicability and patient privacy.

## 7 Conclusion

DR is a leading cause of vision impairment in adults, especially those of working age, resulting from diabetes. This condition develops when high blood sugar levels damage the blood vessels in the retina, the light-sensitive tissue at the back of the eye. DR is classified into two stages: NPDR and PDR. MAs are the early sign of DR detection and are caused by NPDR. Early clinical signs of DR emergence must be identified for better treatment. Since MAs are one of the first visible signs of DR. MAs are small, abnormal swelling or bulging of the retinal blood vessels, so its diagnosis might prevent further vision loss.

In this survey, we systematically review and analyze the published literature on automated detection and segmentation of MAs, aiming to help ophthalmologists in early screening and managing DR. Section 2 displays publicly available datasets with image-level and pixel-level MA annotations. Benchmarking is essential for accurately evaluating the effectiveness of early detection systems for DR. Common evaluation metrics for benchmarking the performance of MA recognition are shown in Section 3.

MA recognition approaches can be broadly classified into two main categories: image processing-based methods and deep learning-based methods. DR cannot be appropriately diagnosed without performing MA segmentation. In addition, with the advancement of technology for fundus imaging, many segmentation methods have been proposed for MA segmentation. These methods are discussed in more detail in our survey paper, and we classified them into image processing-based and deep learning-based methods. Sections 4 and 5 comprehensively list the state-of-the-art MA detection and segmentation methods.

Deep learning techniques, particularly CNNs, dominate tasks like identifying MAs in medical imaging. Specialized architectures like U-Net, Faster R-CNN, and YOLO are adapted for detecting MAs. We finally discussed recent trends and potential future directions in this field. The conclusion is that with advancements in machine learning and computer vision, automated MA recognition approaches have shown significant promise.

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**Data Availability** The datasets analyzed and compared during the current study are publicly available :

- DIARETDB : <https://www.kaggle.com/datasets/nguyenhung1903/diaretdb1-standard-diabetic-retinopathy-database>.
- IDRiD : <https://idrid.grand-challenge.org>.
- DDR : <https://www.kaggle.com/datasets/mariaherrerot/ddrdataset>.
- APTOS 2019 Kaggle : <https://www.kaggle.com/competitions/aptos2019-blindness-detection>.
- ROC : <http://webeye.ophth.uiowa.edu/ROC/>.
- e-ophtha : <https://www.adcis.net/en/third-party/e-ophtha/>.
- FGADR : <https://csyizhou.github.io/FGADR/>.

## Declaration

**Conflicts of interest** We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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